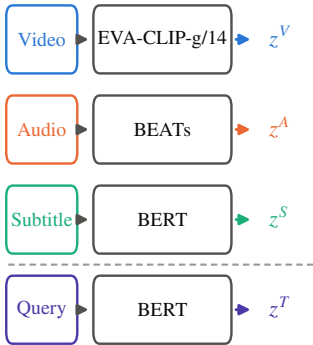


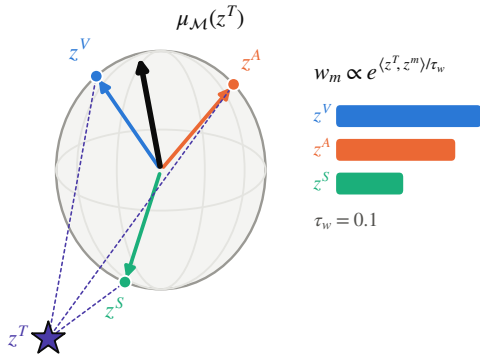
(a) Encoding



query side: never aggregated

VAST backbone frozen
LoRA $r = 8$ ($\alpha = 16$) on q, v

(b) Query-conditioned aggregation



$$s = \langle z^T, \mu_{\mathcal{M}}(z^T) \rangle \in [-1, 1] \quad \text{for any } \mathcal{M} \subseteq \{V, A, S\}$$

a missing modality is one fewer term

(c) Training

